Answer Key - Science - K-12 Level Assessment 13 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** D
   Options: A) a hand lens with 1 X magnification; B) a microscope with 40 X magnification; C) a light microscope with 100 X magnification; D) a dissecting microscope with 4 X magnification
   Note: A dissecting microscope with 4 X magnification is the best tool for observing different sizes of soil particles because it provides sufficient magnification to examine the physical characteristics of larger particles without losing depth perception, which is essential for understanding their size and structure.

2. **Answer:** D
   Options: A) deposition in water; B) slow cooling and crystallization; C) weathering and erosion; D) exposure to heat and high pressure
   Note: The transformation of igneous rock into metamorphic rock occurs through the process of metamorphism, which involves the application of heat and high pressure that alters the mineral composition and structure of the rock without melting it.

3. **Answer:** C
   Options: A) 1; B) 2; C) 3; D) 4
   Note: Three basic units of information, known as nucleotides, are required to form a codon, which encodes a single amino acid in the genetic code.

4. **Answer:** C
   Options: A) Pressure has been converted to light.; B) Force has been converted to light.; C) Closing an electric circuit has produced light.; D) Opening an electric circuit has produced light.
   Note: Closing an electric circuit allows electrical current to flow through the clock's light source, resulting in the illumination of the time display.

5. **Answer:** B
   Options: A) acquired trait to gain an inherited trait.; B) inherited trait to gain an acquired trait.; C) acquired trait to gain another acquired trait.; D) inherited trait to gain another inherited trait.
   Note: The correct answer is based on the understanding that the mouse's sense of smell, an inherited trait, enables it to learn and develop the acquired skill of navigating the maze to locate the cheese.

True / False

6. **Answer:** False
   Options: A) True; B) False

Note: Conventional diesel fuel is derived from fossil fuels, which are non-renewable resources, making it not a renewable fuel option for diesel engines.

7. **Answer:** True
   Options: A) True; B) False
   Note: Plant cells contain chloroplasts, which house chlorophyll that captures sunlight and converts it into chemical energy through the process of photosynthesis.

8. **Answer:** False
   Options: A) True; B) False
   Note: A force pushing a rolling ball in the same direction will not cause it to stop; instead, it will increase the ball's speed unless an opposing force, like friction, is greater than the applied force.

9. **Answer:** False
   Options: A) True; B) False
   Note: The statement is false because as a sledder descends a hill, potential energy decreases while kinetic energy increases, demonstrating the conservation of energy where one form is converted into another.

10. **Answer:** True
    Options: A) True; B) False
    Note: Light waves are electromagnetic waves that inherently oscillate in electric and magnetic fields perpendicular to the direction in which the wave travels, confirming that they are transverse waves.

Long Form Response

11. **Answer:** The atomic mass of an atom is primarily determined by the sum of its protons and neutrons, as these particles contribute most of the atom's mass. For an atom with 20 electrons, 21 neutrons, and 20 protons, the atomic mass would be calculated by adding the number of protons and neutrons together. In this case, the atomic mass would be 20 protons + 21 neutrons = 41. Therefore, the atomic mass of this atom is approximately 41 atomic mass units (amu), since electrons have a negligible mass compared to protons and neutrons. This example illustrates how atomic mass reflects the composition of an atom's nucleus.
    Note: correct because it accurately states that the atomic mass is determined by the sum of protons and neutrons, which are the primary contributors to an atom's mass, while noting that electrons have negligible mass in this calculation.

12. **Answer:** A physical change occurs when a substance changes its form but not its chemical composition, such as when ice melts into water. In this example, the solid ice becomes liquid water, but the chemical structure of H_2O remains unchanged. In contrast, a chemical change involves the formation of new substances, as seen when iron rusts. When iron reacts with oxygen and moisture, it forms iron oxide, demonstrating a change in composition. Understanding these differences is significant in everyday life,

as physical changes often involve reversible processes, while chemical changes can lead to new products and irreversible transformations.

Note: The answer correctly distinguishes between physical changes, which involve changes in state without altering chemical composition (like ice melting), and chemical changes, which result in new substances and altered composition (like rusting iron), highlighting their relevance to everyday experiences.

End of Answer Key